

2(a)	resultant force (in any direction) is zero	B1
	resultant torque/moment (about any point) is zero	B1
2(b)(i)	$a = (v - u) / t$ or gradient or $\Delta v / (\Delta)t$	C1
	e.g. $a = (8.8 - 4.6) / (7.0 - 4.0) = 1.4 \text{ m s}^{-2}$	A1
2(b)(ii)	$s = 4.6 \times 4 + [(8.8 + 4.6) / 2] \times 3$	C1
	$= 18.4 + 20.1$	A1
	$= 39 \text{ (38.5) m}$	
2(b)(iii)	$\Delta E = \frac{1}{2} \times 95 [(8.8)^2 - (4.6)^2]$	C1
	$= 3678 - 1005$	A1
	$= 2700 \text{ (2673) J}$	
2(b)(iv)1.	weight = 95×9.81 (= 932 N)	C1
	vertical tension force = $280 \sin 25^\circ$ or $280 \cos 65^\circ$ (=118.3 N)	C1
	$F = 932 + 118$ $= 1100 \text{ (1050) N}$	A1
2(b)(iv)2.	horizontal tension force = $280 \cos 25^\circ$ or $280 \sin 65^\circ$ (= 253.8 N)	C1
	resultant force = 95×1.4 (= 133 N)	C1
	$133 = 253.8 - R$ $R = 120 \text{ (120.8) N}$	A1